

IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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IBM Applied Data Science Capstone: Falcon 9 landing prediction
GitHub: https://github.com/BHobbs-0/ibm-applied-data-science-capstone

Executive Summary

Methodologies

- Collected launch data from the SpaceX REST API and by scraping Wikipedia
- Cleaned the data and created a landing label (Class: 1 landed, 0 did not)
- Explored it with charts, SQL, a Folium map and a Plotly Dash dashboard
- Tuned four classifiers with GridSearchCV and compared them on a test set

Results

- The first stage landed in 60 of 90 Falcon 9 launches (66.7%), rising sharply after 2014
- KSC LC-39A has the most successful launches and the best success rate (76.9%)
- Newer boosters landed 81% of the time at 2,000 to 4,000 kg; 6,000 to 8,000 kg never landed
- Best model: KNN, 83.3% test accuracy; the decision tree overfits (66.7%)

66.7%
First-stage landing rate, 90 launches

76.9%
Success rate at KSC LC-39A

83.3%
Test accuracy of the best model (KNN)

GitHub: https://github.com/BHobbs-0/ibm-applied-data-science-capstone (all notebooks and the Dash app)

Introduction

Project background and context

- SpaceX advertises Falcon 9 launches at about $62 million; other providers charge upwards of $165 million
- Most of the saving comes from landing and reusing the first stage
- Predicting whether the first stage lands lets us estimate a launch's cost, which helps a competitor bid against SpaceX

Questions we want to answer

- Which factors (payload, orbit, site, flight number, booster) affect landing success?
- How has landing success changed over time?
- Can a classifier predict the landing outcome accurately?

$62M

Falcon 9 launch (advertised)

$165M+

Typical launch from other providers

Methodology: Data Collection and Data Wrangling

Data collection: SpaceX REST API

1 GET past launches (JSON) from api.spacexdata.com/v4

2 Decode IDs via the rockets, launchpads, payloads and cores endpoints

3 Keep Falcon 9 only (90 launches); fill missing payload mass with the mean

4 If the API is down (HTTP 522/525), load IBM's prepared copy

Data collection: web scraping

1 requests.get() the Wikipedia launch list (9 June 2021 snapshot)

2 BeautifulSoup parses the HTML tables

3 Helper functions clean each cell (citations, missing links)

4 DataFrame of 121 launches, 11 fields

Data wrangling

1 Missing values: only LandingPad (28.9%), meaning no pad used

2 Count launches by site, orbit and landing outcome

3 5 outcomes mark a failure (False ASDS/Ocean/RTLS, None ASDS, None None)

4 Class label: 1 = landed, 0 = not; success rate 66.7%

GitHub: https://github.com/BHobbs-0/ibm-applied-data-science-capstone (notebooks 01 to 03)

Methodology: EDA and Interactive Visual Analytics

EDA with data visualisation (pandas, seaborn)

- Scatter plots of flight number and payload against launch site and orbit, coloured by Class
- Bar chart of success rate by orbit; line chart of yearly success
- One-hot encoded categorical features (80 columns, float64) for modelling

EDA with SQL (SQLite, %%sql magic)

- 10 queries on SPACEXTABLE (101 rows): DISTINCT, LIKE, SUM, AVG, MIN
- WHERE with ranges, GROUP BY with COUNT, subquery with MAX
- substr() and CASE for months; BETWEEN with ORDER BY DESC

Interactive map with Folium

- Circle and label at each launch site to see locations
- MarkerCluster of green / red markers for launch outcomes
- MousePosition and PolyLines to measure distances to coast, railway, highway and city

Dashboard with Plotly Dash

- Launch site dropdown drives both charts
- Pie chart: successes by site, or success vs. failure for one site
- Payload range slider and payload vs. outcome scatter, coloured by booster version

GitHub: https://github.com/BHobbs-0/ibm-applied-data-science-capstone (notebooks 04 to 06; dashboard/spacex-dash-app.py)

Methodology: Predictive Analysis (Classification)

Prepare
X: one-hot features
Y: Class

→

Standardise
StandardScaler on X

→

Split
80 / 20, random_state 2
(72 train, 18 test)

→

Tune
GridSearchCV, 10-fold CV

→

Evaluate
Test accuracy and confusion matrix

→

Select
Best model

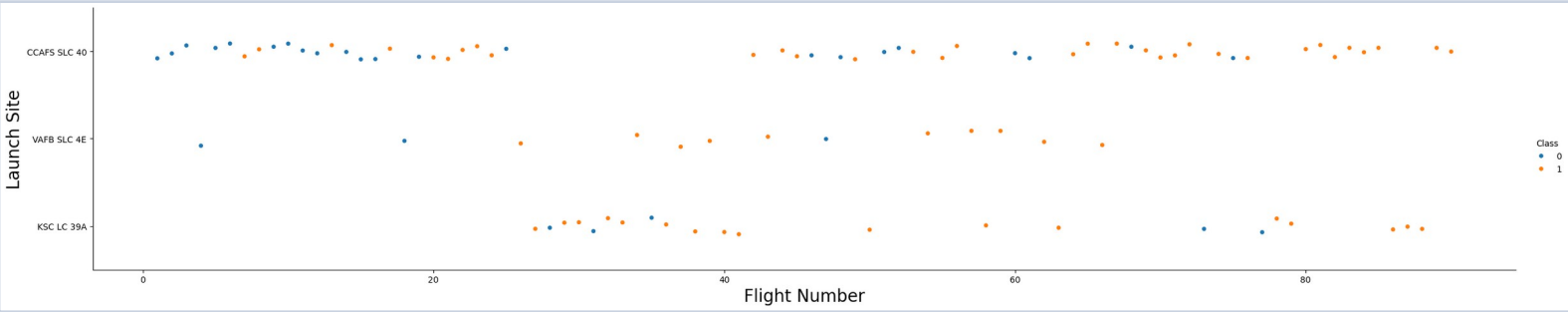
Model	Hyperparameters searched	Best found
Logistic Regression	C: 0.01, 0.1, 1; penalty l2; solver lbfgs	C 0.01
Support Vector Machine	kernel: linear, rbf, poly, sigmoid; C and gamma: 0.001 to 1000	sigmoid, C 1.0, gamma 0.032
Decision Tree	criterion, splitter, max_depth 2 to 18, min_samples_leaf, min_samples_split	entropy, depth 12, random
K Nearest Neighbours	n_neighbors 1 to 10; algorithm; p (1 = Manhattan, 2 = Euclidean)	k 10, p 1

Best model chosen by test accuracy, with validation accuracy as the tie-break.

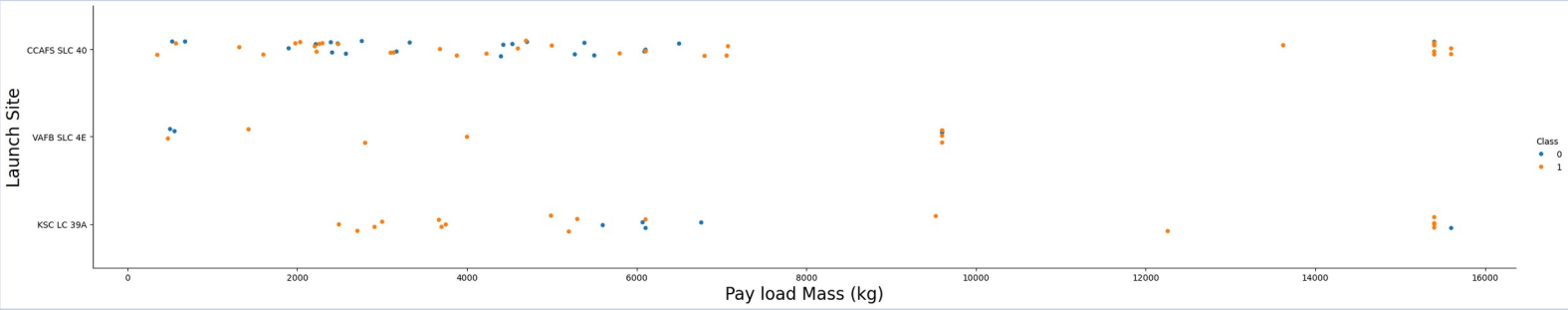
GitHub: https://github.com/BHobbs-0/ibm-applied-data-science-capstone (notebooks/07-machine-learning-prediction.ipynb)

EDA Results: Flight Number and Payload vs. Launch Site

Flight Number vs. Launch Site



Payload Mass vs. Launch Site



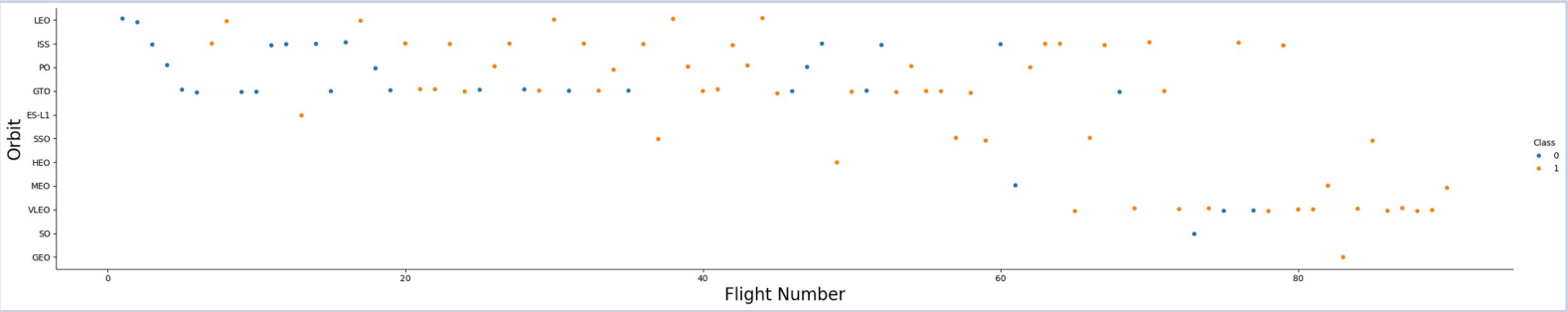
Findings

- CCAFS SLC 40 flew 55 of 90 launches, including the early ones, which mostly failed
- Later flights land far more often at every site
- VAFB SLC 4E launched nothing above 10,000 kg
- The heaviest payloads (over 13,000 kg) mostly landed

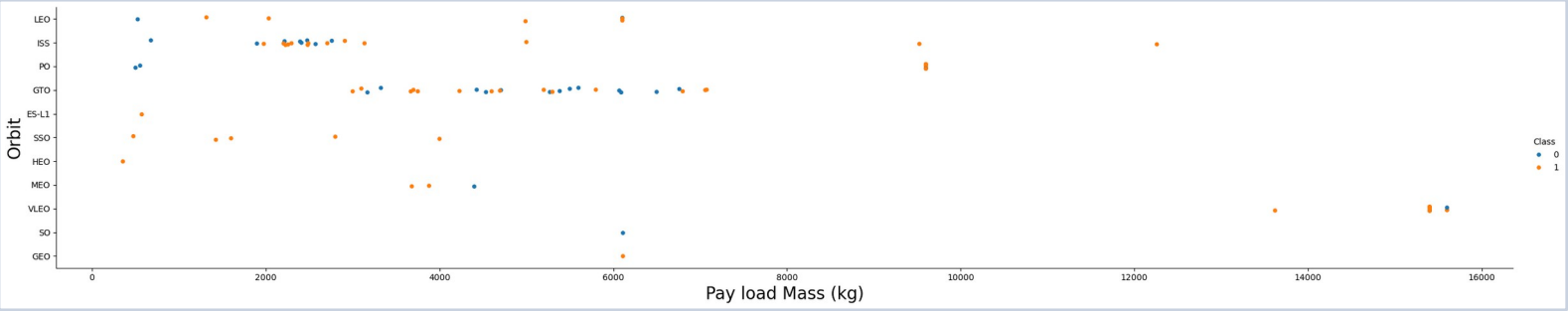
Orange = landed (Class 1), blue = did not land

EDA Results: Flight Number and Payload vs. Orbit Type

Flight Number vs. Orbit



Payload Mass vs. Orbit



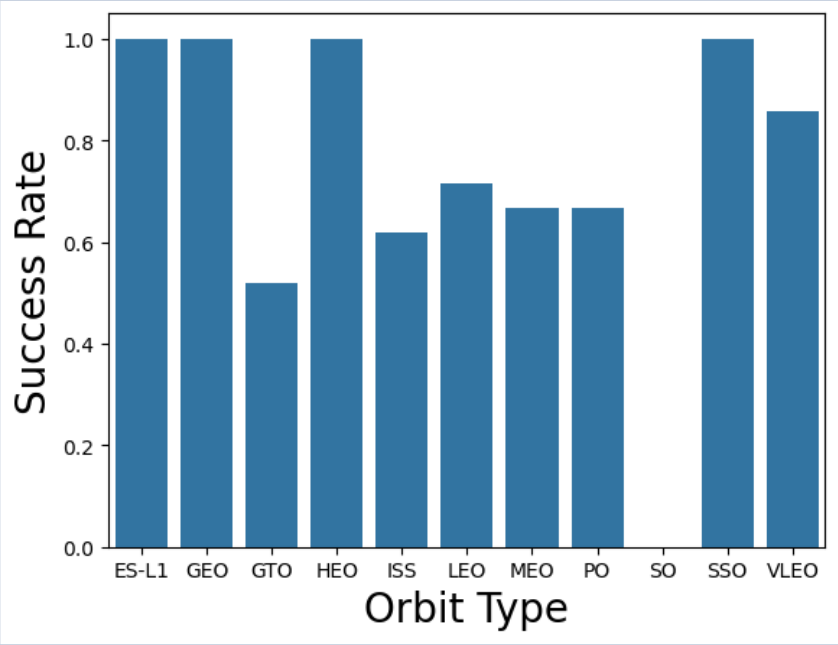
Findings

- LEO success appears related to the number of flights
- GTO shows no clear link between flight number and success
- Heavy payloads land more often for PO, LEO and ISS
- Later VLEO (Starlink) flights carry about 15,000 kg and mostly land

Orange = landed (Class 1), blue = did not land

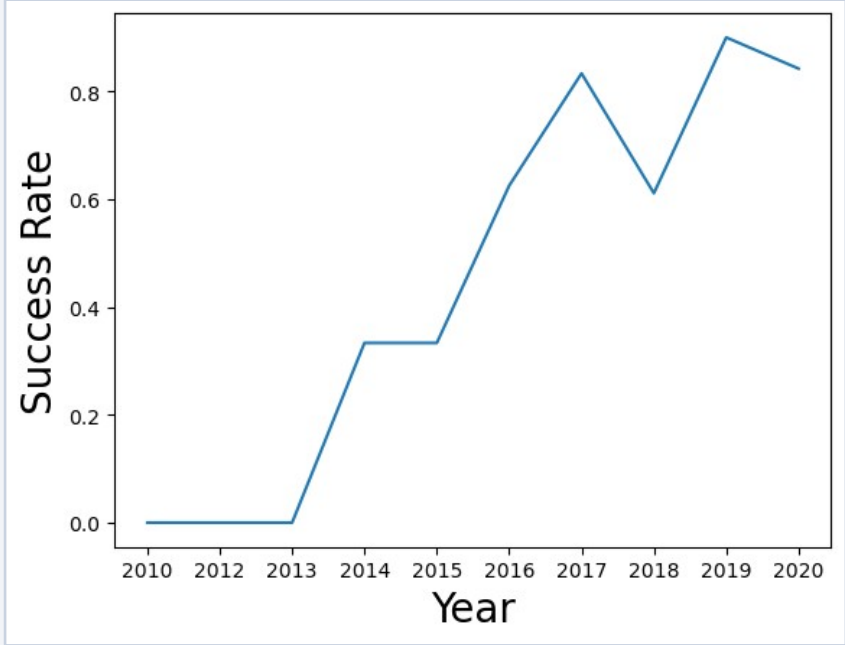
EDA Results: Success Rate by Orbit and Yearly Trend

Success rate by orbit



- ES-L1, GEO, HEO and SSO: 100%, but from only 1, 1, 1 and 5 launches
- GTO, the most common orbit (27), is lowest of the frequent orbits (about 52%)

Launch success yearly trend



- No landings from 2010 to 2013
- Rose to about 83% in 2017, dipped in 2018, about 90% in 2019

EDA with SQL: Queries and Results

#	Task	SQL used	Result
1	Unique launch sites	`SELECT DISTINCT Launch_Site`	CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
2	5 records where site begins 'CCA'	`WHERE Launch_Site LIKE 'CCA%' LIMIT 5`	5 early F9 v1.0 launches from CCAFS LC-40 (2010 to 2013); none landed
3	Total payload, NASA (CRS)	`SUM(PAYLOAD_MASS__KG_) WHERE Customer = 'NASA (CRS)'`	45,596 kg
4	Average payload, F9 v1.1	`AVG(PAYLOAD_MASS__KG_) WHERE Booster_Version = 'F9 v1.1'`	2,928.4 kg
5	First successful ground pad landing	`MIN(Date) WHERE Landing_Outcome = 'Success (ground pad)'`	22 December 2015
6	Drone ship success, 4,000 to 6,000 kg	`WHERE outcome = 'Success (drone ship)' AND mass > 4000 AND < 6000`	F9 FT B1022, B1026, B1021.2, B1031.2
7	Successful and failed mission outcomes	`COUNT(*) GROUP BY Mission_Outcome`	100 successes, 1 failure (in flight)
8	Boosters with the maximum payload	`WHERE mass = (SELECT MAX(PAYLOAD_MASS__KG_) ...)`	12 Block 5 boosters, e.g. B1048.4, B1049.4, B1051.3, B1060.2
9	2015 drone ship failures by month	`CASE substr(Date, 6, 2) ...; substr(Date, 0, 5) = '2015'`	January: F9 v1.1 B1012; April: F9 v1.1 B1015 (both CCAFS LC-40)
10	Rank landing outcomes, 2010-06-04 to 2017-03-20	`BETWEEN dates GROUP BY outcome ORDER BY COUNT DESC`	No attempt 10; drone ship success 5, failure 5; ground pad success 3; ...

GitHub: https://github.com/BHobbs-0/ibm-applied-data-science-capstone (notebooks/04-eda-sql.ipynb)

Folium Map Results: Launch Sites and Their Proximities



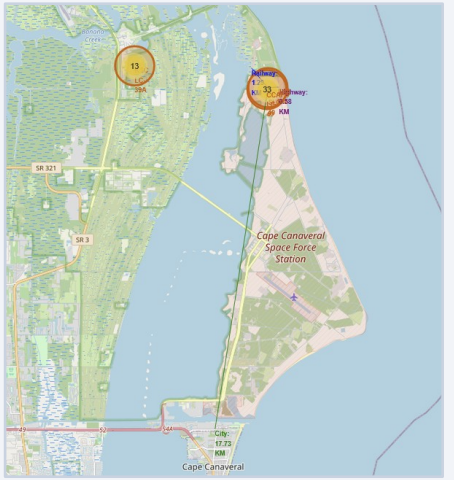
All sites are on the US coast

- Three in Florida (about 28.6°N), close to the equator
- VAFB in California, for polar launches over the Pacific



Launch outcomes (green = success, red = failure)

- KSC LC-39A is mostly green: 10 of 13 landed
- CCAFS LC-40 mostly red: 7 of 26 (early flights)



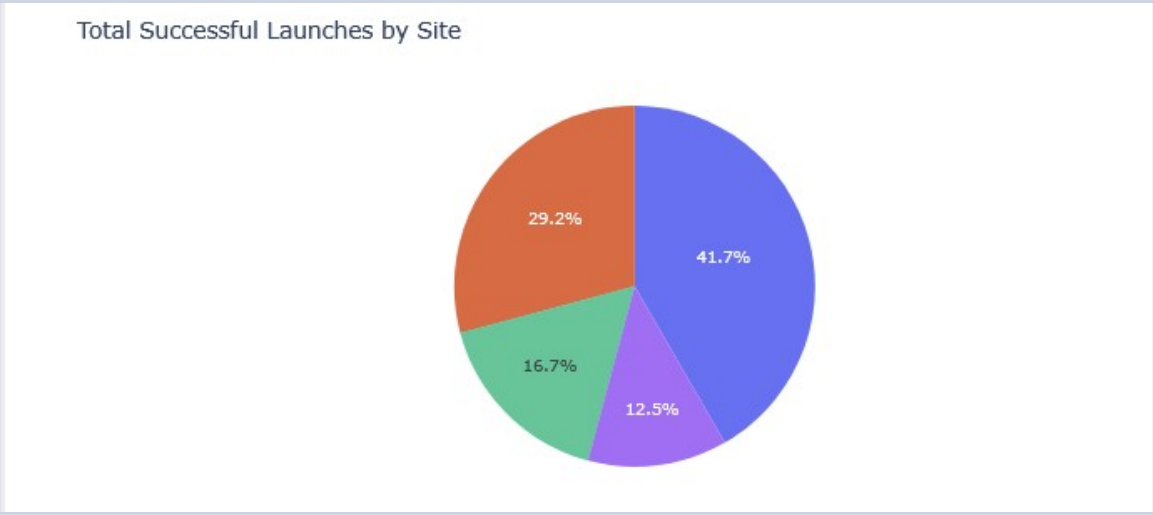
CCAFS SLC-40 proximities

- Coast 0.51 km, highway 0.58 km, railway 1.29 km
- City 17.73 km away, for safety

GitHub: https://github.com/BHobbs-0/ibm-applied-data-science-capstone (notebooks/06-interactive-map-folium.ipynb)

Plotly Dash Results: Launch Success by Site

All Sites: share of successful launches



KSC LC-39A 41.7%

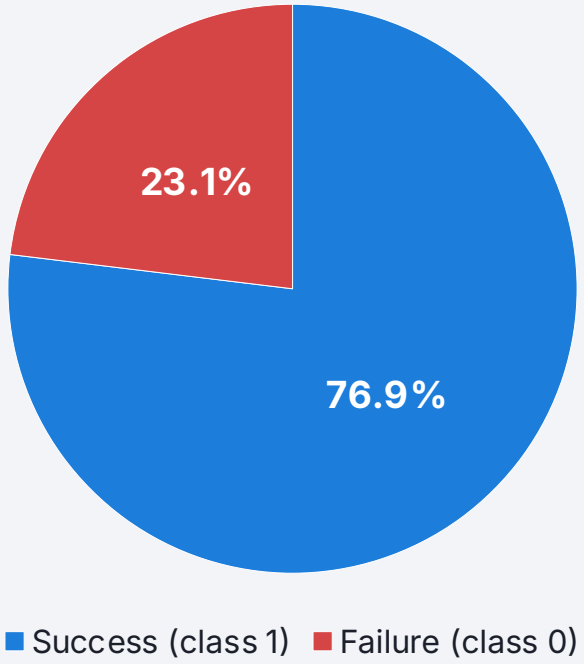
CCAFS LC-40 29.2%

VAFB SLC-4E 16.7%

CCAFS SLC-40 12.5%

KSC LC-39A has the largest share of all successful launches.

KSC LC-39A selected: highest success ratio



76.9% success (10 of 13); next best is CCAFS SLC-40 at 42.9% (3 of 7).

Plotly Dash Results: Payload vs. Launch Outcome



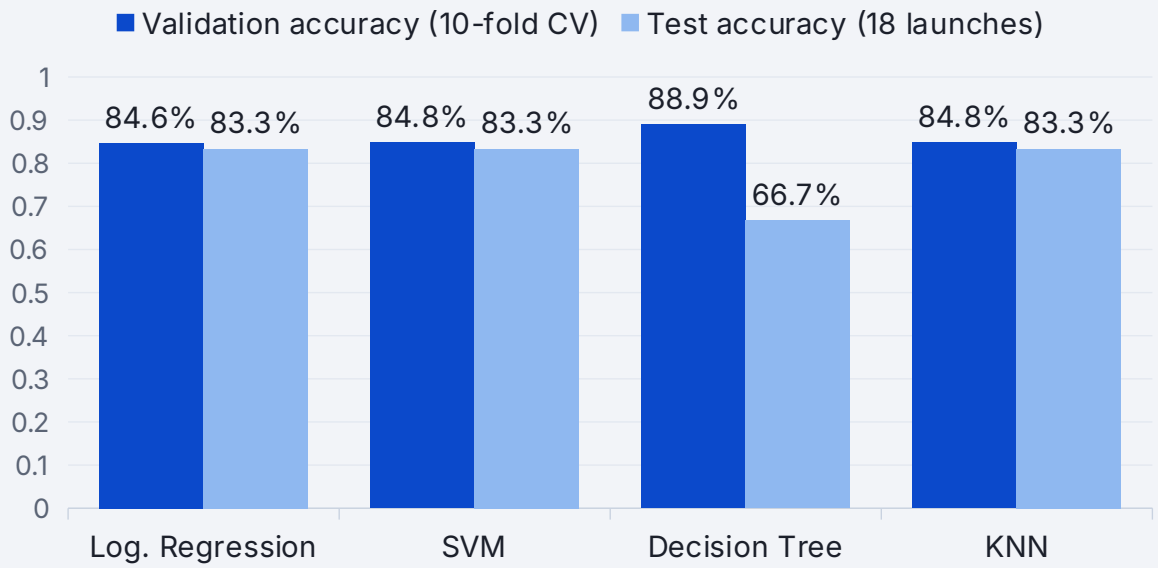
All Sites, payload slider 0 to 9,600 kg; colour = booster version

Payload range	Success rate
0 to 2,000 kg	23% (3 of 13)
2,000 to 4,000 kg	62% (13 of 21)
4,000 to 6,000 kg	38% (5 of 13)
6,000 to 8,000 kg	0% (0 of 4)
8,000 to 10,000 kg	60% (3 of 5)

- 2,000 to 4,000 kg has the highest success rate; 6,000 to 8,000 kg the lowest
- FT boosters have the best record (67% of 24); v1.1 landed 1 of 15

GitHub: https://github.com/BHobbs-0/ibm-applied-data-science-capstone (dashboard/spacex-dash-app.py)

Predictive Analysis Results: Accuracy and Confusion Matrix



Model	Validation	Test	False pos.
Logistic Regression	84.6%	83.3%	3
SVM	84.8%	83.3%	3
Decision Tree	88.9%	66.7%	6
KNN (best)	84.8%	83.3%	3

Best model: KNN

83.3% test accuracy (15 of 18), 84.8% validation

KNN confusion matrix (rows: true, columns: predicted)

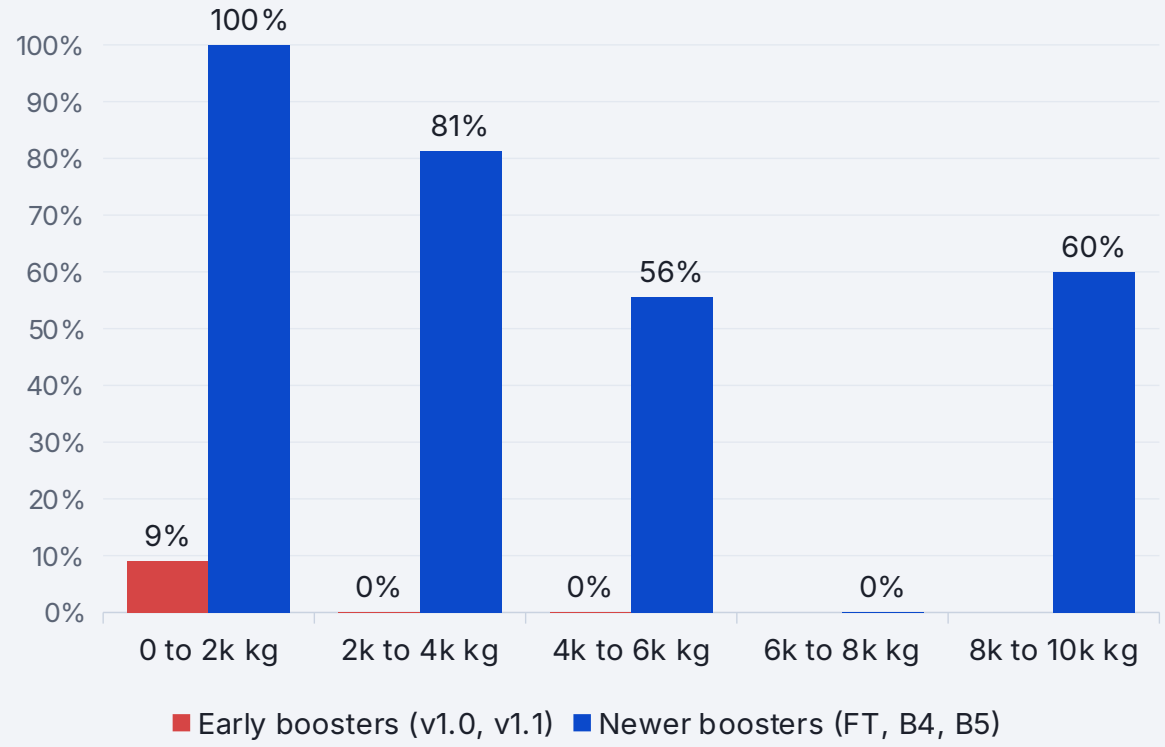
	Did not land	Landed
Did not land	3	3
Landed	0	12

Why KNN: joint-highest test and validation accuracy (with SVM); the notebook's comparison selects it.

Weakness: 3 false positives, so it is too optimistic about landings. The decision tree overfits (best validation, worst test).

Innovative Insights: Booster Age, Payload and a Competing Bid

Landing success by payload band and booster age



From spacex_launch_dash.csv (56 launches), beyond the lab tasks

What the data shows

- Early boosters landed only 1 of 20 times; they flew most light payloads, so "light payloads fail" is really booster age
- Newer boosters land 81% at 2,000 to 4,000 kg; 6,000 to 8,000 kg failed every time

What a competitor should do

- Bid on heavy GTO missions, where SpaceX is least likely to land and reuse the booster
- Run each tender through the KNN model; a predicted non-landing means SpaceX's cost is closer to full price
- Act soon: success rose from 0% (2013) to about 90% (2019)

Conclusions

- Landing success improved with experience: none before 2014, about 90% by 2019
- KSC LC-39A has the most successful launches (41.7% of all successes) and the best success rate (76.9%)
- Payload matters: 2,000 to 4,000 kg landed most often, 6,000 to 8,000 kg never did
- Booster age matters as much: early v1.0 and v1.1 boosters landed 1 of 20 times
- GTO, the most common orbit, is the hardest to land; ES-L1, GEO, HEO and SSO always landed but are rare
- KNN is the best model: 83.3% test accuracy (tied with SVM and logistic regression) and 84.8% validation
- The decision tree overfits; with only 18 test launches, more data would make the comparison more reliable

KNN

Best model: 83.3% test accuracy (15 of 18)

76.9%

Success rate at KSC LC-39A, the best site

Appendix: GitHub Repository and Creative Additions

GitHub repository: https://github.com/BHobbs-0/ibm-applied-data-science-capstone

Lab	File in the repository
Data collection (SpaceX API)	notebooks/01-data-collection-api.ipynb
Data collection (web scraping)	notebooks/02-data-collection-webscraping.ipynb
Data wrangling	notebooks/03-data-wrangling.ipynb
EDA with SQL	notebooks/04-eda-sql.ipynb
EDA with data visualisation	notebooks/05-eda-visualisation.ipynb
Interactive map with Folium	notebooks/06-interactive-map-folium.ipynb
Plotly Dash dashboard	dashboard/spacex-dash-app.py
Predictive analysis	notebooks/07-machine-learning-prediction.ipynb

Creativity beyond the template

- Flowcharts for every method
- Stat callouts and native charts
- All 10 SQL results in one table
- API outage fallback (HTTP 522 / 525)
- Booster age vs. payload analysis
- Competing-bid recommendations